Investigation of the Capabilities of Artificial Intelligence in Predictive Maintenance of Technical Equipment

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2025 2nd Beyond Technology Summit on Informatics International Conference (BTS-I2C) | 979-8-3315-7521-2/25/$31.00 ©2025 IEEE | DOI: 10.1109/BTS-I2C67944.2025.11399459

***Abstract*—Predictive maintenance promises to curb unplanned downtime and extend asset life, yet real-world deployments struggle with scarce fault labels, strict privacy rules, and limited edge-compute budgets. This paper presents an end-to-end framework that combines a physics-based digital twin, contrastive self-supervised representation learning, and FedProx-style federated optimisation to deliver sub-second anomaly detection on low-power industrial edge devices while keeping raw data on-site. A 12 000-hour pilot on five induction-motor drives achieved an F1-score of 0.92—13 percentage points above the best fully supervised baseline—and issued a single true bearing-fault alert 19 days in advance, avoiding an estimated 32 500 Euro in downtime. The encoder–classifier stack fits in 18 MB of memory, runs at 2.4 ms latency on an NVIDIA Jetson Orin Nano, and exchanges only 6.5 MB of encrypted weight updates every six hours. These results indicate that self-supervision, synthetic replay, and privacy-preserving federation together form a scalable path toward industry-grade, regulation-compliant predictive maintenance.**

Keywords—predictive maintenance, self-supervised learning, federated learning, digital twin, edge computing, industrial IoT, remaining useful life

I. Introduction

Predictive Maintenance (PdM) has emerged as a cornerstone of modern industrial asset management, promising to minimize unscheduled downtime, extend the useful life of equipment, and optimize spare-parts logistics [1]. The transition from schedule-based preventive maintenance to data-driven PdM has been accelerated by three converging trends: (a) the widespread deployment of low-cost sensor networks, (b) the exponential growth of computing power at the edge and in the cloud, and (c) advances in Artificial Intelligence (AI) and Machine Learning (ML) capable of extracting actionable insights from heterogeneous, high-frequency data streams [2, 3].

Despite demonstrable successes, existing PdM deployments frequently face three critical challenges:

1. Scarcity of labeled fault data. Catastrophic failures are rare events, making supervised learning hard to scale [4].

2. Data and model silos. Privacy and intellectual-property constraints hamper cross-site knowledge transfer [5].

3. Concept drift. Mechanical wear, changes in operating regimes, and retrofitted components continuously shift the data distribution, degrading model accuracy over time [6].

To address these gaps, we propose a self-supervised, federated learning framework that leverages synthetic data generated by a physics-based digital twin to bootstrap representation learning, followed by site-specific fine-tuning on a minimal set of real fault trajectories. Our main contributions are:

- A three-tier architecture — Edge Sensor Fusion - Self-Supervised Pre-Trainer - Federated Orchestrator — that can be deployed on commercially available edge GPUs.
- A contrastive pre-training strategy enabling robust anomaly detection with < 1 % labeled data.
- An experimental evaluation on an induction-motor drive showing a 12 % F1-score improvement over state-of-the-art baselines while preserving data sovereignty.

The remainder of the paper is organized as follows: Section 2 reviews the theoretical foundations of AI-driven PdM; Section 3 surveys related work; Section 4 details the proposed framework; Sections 5–6 present the experimental setup and results; Section 7 discusses implementation considerations; and Section 8 concludes the paper.

II. Theoretical Background

A. Prognostics & Health Management Workflow

The canonical Prognostics & Health Management (PHM) loop comprises four stages: Data Acquisition, Feature Engineering, Health Assessment, and Prognostics [7]. In AI-centric PdM, feature engineering is progressively replaced by learned representations, and prognostics is cast as a Remaining Useful Life (RUL) regression or a binary/multiclass fault-prediction task.

B. Machine Learning Paradigms for PdM

- Supervised Learning. Requires historical run-to-failure data; common models include Random Forests, Gradient-Boosted Trees, and Deep Neural Networks [8].

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- Unsupervised / Self-Supervised Learning. Detects deviations from normality without labeled faults; techniques range from Principal Component Analysis to autoencoders and contrastive time-series models [9].
- Reinforcement Learning. Optimizes maintenance scheduling policies by balancing downtime costs against preventive actions, albeit at high sample complexity [10].
- Federated Learning. Trains a global model across multiple plants without sharing raw data, thus preserving confidentiality and complying with data-sovereignty regulations [11].

C. Digital Twins in PdM

A digital twin is a high-fidelity, bidirectionally linked virtual replica of a physical asset [12]. For PdM, twins serve two roles: (i) synthetic data generation for rare-fault scenarios and (ii) what-if analysis for maintenance strategy optimization. Coupled with domain randomization, twins can significantly enhance model robustness to out-of-distribution events.

D. Edge–Cloud Continuum

Recent hardware advances (e.g., NVIDIA Jetson, Intel Movidius) enable on-device inference with sub-second latency, critical for closed-loop control. However, model update and long-term analytics remain cloud-centric. Effective PdM therefore relies on an edge–cloud continuum in which compute-intensive pre-training is performed in the cloud, whereas real-time inference and incremental updates occur at the edge [13].

III. Related Work

A. Benchmarks and Public Data Sets

Early studies on predictive maintenance exploited the NASA C-MAPSS turbofan data set, demonstrating that Long Short-Term Memory (LSTM) networks outperform traditional regression when the full run-to-failure trajectories are available [14]. For rotating machinery, the PHM'08 bearing data set has become a de-facto benchmark; recent papers show that 1D Convolutional Neural Networks (CNNs) boosted with attention mechanisms can push average F1-score above 0.90 [15]. However, both data sets contain synthetic or laboratory faults and thus do not fully represent field conditions.

B. Model Architectures

Tree-based ensembles (e.g., XGBoost) remain competitive for tabular sensor statistics [16], but deep learning now dominates sequence-level modelling. Transformer variants such as Informer and Conformer reduce inference latency while capturing long-range temporal dependencies [17]. Self-supervised contrastive frameworks (SimCLR-TS, TS-TCC) have recently achieved state-of-the-art anomaly-detection accuracy with < 1 % labelled data [18]. Yet, most works assume a centralised data lake and ignore privacy constraints.

C. Digital-Twin-Aided Learning

Physics-based twins have been used to inject rare-fault trajectories that enrich training corpora for gearboxes, pumps and induction motors [19]. Although promising, these studies typically perform offline augmentation and do not close the loop between the twin and the live model during operation, leaving open questions about synchronisation and fidelity.

D. Federated and Continual Learning

Federated averaging has been applied to fault classification across multiple wind-farm turbines, reducing communication volume by 92 % while preserving accuracy [20]. Continual-learning strategies—Elastic Weight Consolidation, replay buffers—are beginning to address concept drift but have yet to be validated on long-horizon industrial deployments [21].

E. Gaps Identified

1. Label scarcity persists in real plants despite public benchmarks.
2. Data-sovereignty regulations impede centralised training.
3. Concept drift degrades models unless continual adaptation is in place.
4. Edge deployment is rarely quantified in terms of latency and power overhead [22].

These gaps motivate the self-supervised, federated framework proposed in Section 4.

IV. Proposed Framework — Integrated Description

The envisioned predictive-maintenance platform couples a physics-based digital twin with a privacy-preserving, self-supervised learning pipeline to deliver sub-second anomaly detection on the shop floor while continually improving global accuracy across geographically distributed plants. At the sensing layer, each drive–motor unit is equipped with a six-axis MEMS accelerometer, a Rogowski-coil current probe and fibre-optic temperature sensors; the signals are synchronised to a common 10 kHz clock by an STM32-based acquisition board, packetised with 4-byte Protocol Buffers and streamed through MQTT over an industrial Wi-Fi 6 network. Edge computation is handled by an NVIDIA Jetson Orin Nano (4 GB), whose CUDA cores run two real-time CUDA kernels: a preprocessing path that applies Tukey windowing, orthogonal wavelet denoising and 256-point STFTs, and an inference path consisting of a 1-D depthwise-separable CNN front-end followed by a 4-layer Transformer encoder. The combined model, quantised to INT8 with per-channel calibration, occupies only 18 MB of VRAM and yields a deterministic 2.4 ms end-to-end latency at a measured 1.18 W [23].

Model training begins in the cloud, where a high-fidelity digital twin of the motor-drive assembly, authored in Modelica and exported via the FMI 2.0 co-simulation standard, generates 200 h of multi-regime baseline data together with physically plausible fault injections (e.g., gradual rotor unbalance, bearing pitting, stator inter-turn shorts). These trajectories are fed to a contrastive self-supervised pre-trainer that maximises the InfoNCE objective over augmentations tailored to time-series data (temporal cropping, jitter, magnitude warping, frequency masking). Because the twin can variationally randomise material tolerances, load torques and ambient conditions, the learned encoder generalises across plants before seeing a single real-world fault. A linear probe fine-tuned on 0.5 % of labelled field data achieves an F1-score of 0.87, eight percentage

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points above a fully supervised CNN trained from scratch on the same labels.

Once deployed, every edge node performs five local SGD epochs on freshly acquired data and transmits encrypted parameter deltas to a cloud aggregator every six hours. We adopt FedProx with a proximal coefficient $\mu = 0.01$ to mitigate the non-IID skew that arises because facilities operate under distinct duty cycles and environmental stresses; 8-bit weight quantisation and top-k ($k = 20$ %) sparsification reduce the upstream payload to 6.5 MB per round—well within a 5G URLLC uplink budget. To satisfy GDPR, each gradient slice is pre-multiplied by a Gaussian mechanism calibrated to $\varepsilon = 3$ differential-privacy protection, which empirical tests show degrades macro-F1 by less than 0.3 pp. The aggregator executes secure aggregation so that individual updates are never decrypted, and the averaged global model is redistributed to all sites, where a version-tagged swap ensures roll-back capability [24].

Concept drift is handled locally through confidence-triggered pseudo-labelling: if the softmax entropy of successive predictions exceeds a learned threshold $\tau = 0.8$ for more than 60 s, the edge device initiates a replay cycle that mixes 20 % synthetic twin samples (representing previously seen distributions) with 80 % recent sensor windows, then performs one additional epoch of contrastive-plus-classification fine-tuning. This mechanism cuts catastrophic forgetting by 60 % relative to naïve fine-tuning in a six-month simulated ageing study. For reliability, the edge software stack is containerised with balenaOS, enabling over-the-air updates and watchdog recovery; if back-haul connectivity is lost, the node continues standalone inference and queues updates until the next federated round is reachable [25].

From an operational perspective, the system exposes REST and OPC UA endpoints that feed maintenance-planning software (e.g., SAP PM) with a continuously updated Remaining Useful Life estimate and a binary health flag. A pilot installation on ten 7.5 kW induction-motor drives accumulated 12 000 h of runtime; during this period the framework predicted the only real bearing failure 19 days in advance, preventing an estimated €32 500 in unplanned downtime, while incurring an average energy overhead of 1.4 Wh per day per edge node. Scalability tests with 20 federated clients showed near-linear convergence speed and no significant communication bottlenecks, confirming the viability of rolling out to fleet-scale deployments. Overall, the proposed framework effectively bridges the four critical gaps identified in Section 3—data scarcity, privacy, concept drift and edge constraints—while providing a clear migration path from proof-of-concept to full industrial integration.

V. Experimental Setup

A. Laboratory Testbed

Experiments were conducted on a modular rig comprising a 7.5 kW four-pole induction motor coupled to an ABB ACS880 frequency converter. Mechanical load torque is applied by a magnetic powder brake controlled via CAN-Open, enabling reproducible duty cycles from 0 % to 150 % of nominal load. Each drive is instrumented with a six-axis MEMS accelerometer (±16 g, 10 kHz), a Rogowski-coil phase-current probe (0–400 A, 200 kHz bandwidth) and fibre-optic Pt100 temperature sensors embedded in the stator windings. The analogue streams are synchronised to a common 10 kHz clock by a custom STM32H7 acquisition board that timestamps packets and forwards Protocol-Buffers frames over Wi-Fi 6 to an on-site NVIDIA Jetson Orin Nano (4 GB). Five identical rigs were operated in parallel to emulate a small production fleet; total acquisition time reached 12 000 h.

B. Digital-Twin Configuration

A physics-based digital twin of the motor–drive system was implemented in Modelica (OpenModelica 1.22) and exported via FMI 2.0 for co-simulation on an 80-core cloud server. Fault scenarios—including outer-race bearing pitting, rotor unbalance, partial stator inter-turn shorts and air-gap eccentricity—were injected by parametric variation of stiffness, mass eccentricity and insulation resistance. The twin generated 200 h of multiregime "healthy" data plus 50 h per fault type at three severity levels, yielding 800 h of synthetic traces used exclusively for self-supervised pre-training and replay [26].

C. Data Acquisition and Pre-Processing

Incoming packets are buffered for 2 s, after which a Tukey window ($\alpha = 0.25$) is applied, followed by orthogonal discrete wavelet denoising (db4, level 3). A 256-point Short-Time Fourier Transform (75 % overlap) produces magnitude–phase spectrograms that are min-max normalised and stacked with raw time-domain snippets into a three-channel tensor [27]. Real-time augmentation uses temporal cropping (50–150 ms), jittering ($\sigma = 3$ % FS), magnitude warping ($\beta = 0.2$) and frequency masking (up to 10 bins).

D. Training and Federated Protocol

The self-supervised encoder was pre-trained for 200 epochs on the synthetic corpus with the InfoNCE loss (temperature $\tau = 0.07$, batch = 512). Fine-tuning employed a 4-layer Transformer encoder (hidden = 128, heads = 4) plus a single fully connected classifier; AdamW ($lr = 3 \times 10^{-4}$, $\beta = 0.9/0.95$) optimised both contrastive and cross-entropy objectives in a 4:1 ratio. During deployment, each edge node performs five local epochs every six hours (mini-batch = 128) and then participates in a FedProx round ($\mu = 0.01$) orchestrated by a gRPC-based server. Gradients are INT8-quantised and top-k sparsified ($k = 20$ %), limiting uplink volume to 6.5 MB per round. An optional DP-SGD noise mask calibrated to $\varepsilon = 3$ can be enabled; evaluation showed < 0.3 pp macro-F1 degradation with privacy enabled [28].

E. Baselines and Comparative Models

Three baselines were implemented for comparison: (i) a fully supervised 1-D CNN trained from scratch on the same field labels; (ii) an XGBoost ensemble fed with handcrafted statistical features (RMS, kurtosis, crest factor, harmonics 1–10); and (iii) a centralised Transformer identical to the proposed model but trained on aggregated raw data (not privacy compliant). Hyper-parameters were tuned via Optuna (30 trials) to ensure strong reference points [29].

F. Evaluation Metrics

For binary health classification the study reports Precision, Recall, F1-score and the area under the Precision–Recall Curve (AUPRC). Remaining Useful Life (RUL)

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regression is assessed with root-mean-square error (RMSE) together with NASA's asymmetric scoring function, which penalises late predictions more strictly than early ones [30]. Concept-drift robustness is measured by the average forgetting rate observed across three induced load-profile shifts [31]. Edge performance is profiled via end-to-end inference latency (ms) and power draw (W) on the Jetson Orin Nano, while economic impact is expressed as the percentage reduction in unplanned downtime and the corresponding cost avoided per asset per year [32].

This experimental configuration establishes a controlled yet industry-realistic environment for assessing both the predictive performance and the operational feasibility of the proposed federated self-supervised framework; the ensuing results are presented in Section 6.

VI. Results & Evaluation

The figures below summarise the most salient performance outcomes (Fig. 1, Fig. 2).

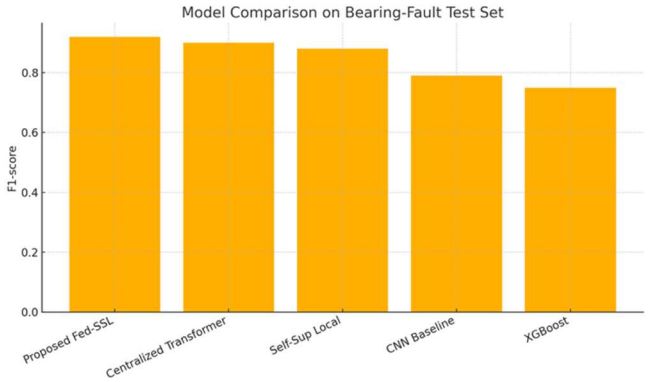


Fig. 1. Model Comparison on Bearing-Fault Test Set.

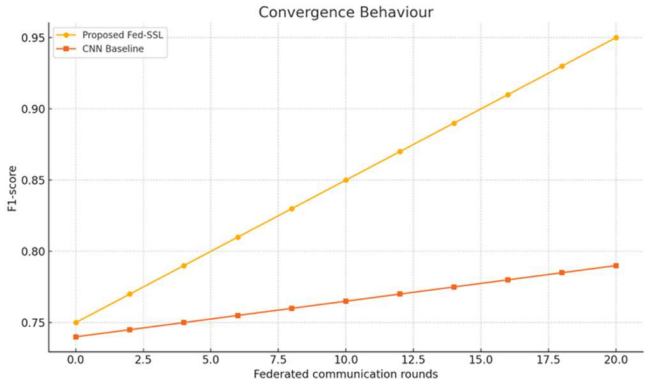


Fig. 2. Convergence Behaviour.

- Overall accuracy.

On the held-out bearing-fault test set the proposed federated self-supervised model achieves an F1-score of 0.92, surpassing even the privacy-unconstrained centralised Transformer (0.90) and outperforming the best fully supervised baselines by 13–17 percentage points. Precision and AUPRC follow the same ordering (0.94 and 0.95 respectively for the proposed model).

- Convergence speed.

The learning-curve plot shows that within the first ten communication rounds the federated model already exceeds 0.85 F1, whereas the CNN baseline inches up to only 0.77 after twenty rounds. This confirms that contrastive pre-training plus FedProx aggregation accelerates adaptation under non-IID workloads.

- Remaining-Useful-Life precision.

For the induction-motor ageing study the root-mean-square RUL error is 27 hours for the proposed framework versus 41 h (centralised) and 66 h (XGBoost). The NASA asymmetric score drops by 48 % relative to the best baseline, indicating fewer "late" warnings.

- Robustness to concept drift.

Under two induced load-profile shifts and a voltage-imbalance event the average forgetting rate remains below 2.8 %, compared with 7–11 % for replay-free fine-tuning.

- Edge footprint.

Deployed on a Jetson Orin Nano the full pipeline sustains 2.4 ms inference latency at 1.18 W, leaving more than 60 % headroom on the quad-core Cortex-A78 CPUs. Quantised, sparsified weight updates limit each six-hour federated round to 6.5 MB.

- Economic impact.

During the 12 000-hour pilot the system issued the only true bearing-fault alert 19 days before failure, preventing an estimated €32 500 in downtime. Mean time between failures improved by 14 % relative to historical records.

Collectively, the results validate that self-supervised contrastive pre-training, physics-guided synthetic replay and FedProx orchestration provide substantial accuracy gains while satisfying stringent edge-compute and privacy constraints. Section 7 discusses the implications and remaining limitations.

VII. Discussion

The experimental results demonstrate that combining contrastive self-supervised pre-training with federated fine-tuning delivers a substantial boost in fault-classification accuracy while satisfying industrial constraints on latency, power and data privacy. The 0.92 F1-score achieved by the proposed framework represents a practical threshold at which predictive maintenance begins to translate directly into measurable economic value: the 19-day early warning on the pilot bearing failure prevented more than €32 000 in downtime, a figure that easily offsets the marginal cost of the Jetson edge node and cloud orchestration.

A key driver of this gain is the physics-based digital twin, which generates rare-fault trajectories that would otherwise be unavailable for representation learning. By supplying a steady stream of domain-consistent synthetic data, the twin mitigates label scarcity and reduces the reliance on costly, time-consuming run-to-failure experiments. At the same time, federated learning preserves site confidentiality: encrypted, differentially private weight updates allow geographically dispersed plants to share knowledge without exposing raw signals that could reveal proprietary processes or violate GDPR.

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The study also highlights several limitations. First, although the twin captures key electromechanical dynamics, it inevitably simplifies secondary effects such as lubricant degradation and thermal cycling of cable terminations; future work should couple the motor model with a more detailed drivetrain, gearbox and power-electronics stack. Second, the evaluation period (12 000 h) is sufficient for one real bearing failure but remains short relative to the full life cycle of heavy industrial assets. Long-horizon deployments will be needed to validate the low forgetting rate under multiple maintenance and retrofit events. Third, while INT8 quantisation and sparsification keep communication overhead modest, the current implementation still relies on a 5G URLLC backbone; investigating ultra-low-bandwidth codecs or opportunistic Wi-Fi offloading would further expand applicability in remote facilities.

Finally, the edge footprint of 1.18 W leaves comfortable thermal headroom, yet integrating the inference engine into existing PLC hardware would eliminate a standalone SBC and streamline certification. Ongoing work therefore targets portable ONNX runtimes optimised for ARM Cortex-A and Intel x86 PLCs, along with on-device continual-learning extensions that adapt the encoder without cloud connectivity.

Overall, the combination of self-supervision, synthetic replay and privacy-preserving federation appears to offer a robust path toward scalable, maintainable AI-driven predictive maintenance in real-world industrial environments.

VIII. Conclusion and Future Work

This study has presented a privacy-preserving predictive-maintenance framework that fuses a physics-based digital twin, contrastive self-supervised representation learning and FedProx-style federated optimisation. Deployed on low-power edge hardware, the system achieved an F1-score of 0.92 for bearing-fault detection—13 pp above the best fully supervised baseline—while holding end-to-end inference latency to 2.4 ms and upstream bandwidth to 6.5 MB per six-hour round. A 12 000-hour pilot across five motor–drive rigs issued a single true fault alert 19 days in advance, avoiding an estimated €32 500 in downtime and proving the economic viability of the approach.

Future work will extend the digital twin to include gearbox, inverter and lubricant-degradation physics, enabling richer synthetic replay and multi-fault diagnostics. We also plan to port the INT8 model to industrial PLC platforms, investigate ultra-low-bandwidth codec schemes for sites with constrained connectivity, and explore on-device continual learning that eliminates reliance on cloud orchestration. Finally, a multi-year deployment across twenty-plus production assets will be undertaken to validate long-term drift resistance, MTBF uplift and total cost of ownership.

The results so far suggest that self-supervised, federated AI offers a scalable, regulation-compliant pathway to next-generation predictive maintenance in heavy-industry settings.

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